

Projections and Linear Systems

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Design Matrix

Definition 0.1 (Design matrix (model matrix)). In a regression model with n observations and p predictors, the **design matrix** (or **model matrix**) $\mathbf{X}$ is the $n \times p$ matrix whose i -th row is the covariate vector $\tilde{x}_i^\top$ for observation i :

$$\begin{aligned}\mathbf{X} &= \begin{bmatrix} \tilde{x}_1^\top \\ \tilde{x}_2^\top \\ \vdots \\ \tilde{x}_n^\top \end{bmatrix} \\ &= \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}\end{aligned}$$

Remark 0.1 (Products with the design matrix). The product $\mathbf{X}\tilde{\beta}$ collects the values $\tilde{x}_i^\top \tilde{\beta}$ for all n observations into a single $n \times 1$ vector:

$$\mathbf{X}\tilde{\beta} = \begin{bmatrix} \tilde{x}_1^\top \tilde{\beta} \\ \vdots \\ \tilde{x}_n^\top \tilde{\beta} \end{bmatrix}$$

For example, take the rank-2 matrix from [Example 2 in Subspaces and Rank](#) as $\mathbf{X}$, with $n = 3$ observations and $p = 2$ columns: an intercept column and one covariate. With $\tilde{\beta} = (1, 2)$,

$$\begin{aligned}\mathbf{X}\tilde{\beta} &= \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}.\end{aligned}$$

The matrix $\mathbf{X}^\top \mathbf{X}$ is $p \times p$ and symmetric, since

$$\begin{aligned}(\mathbf{X}^\top \mathbf{X})^\top &= \mathbf{X}^\top (\mathbf{X}^\top)^\top \\ &= \mathbf{X}^\top \mathbf{X}\end{aligned}$$

([Theorem 16 in Matrices](#)). When $\mathbf{X}^\top \mathbf{X}$ is invertible, it appears in the OLS estimator $\hat{\tilde{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{y}$.

Theorem 0.1 ($\mathbf{X}^\top \mathbf{X}$ is invertible when $\mathbf{X}$ has full column rank). *If $\mathbf{X}$ is an $n \times p$ matrix with $\text{rank}(\mathbf{X}) = p$ ([Definition 2 in Subspaces and Rank](#)), then the $p \times p$ matrix $\mathbf{X}^\top \mathbf{X}$ is invertible.*

Proof

Proof. Let $\tilde{c}$ be a vector of length p with $\mathbf{X}^\top \mathbf{X} \tilde{c} = \tilde{0}$. Then

$$\begin{aligned}0 &= \tilde{c}^\top \mathbf{X}^\top \mathbf{X} \tilde{c} && \text{(multiply } \mathbf{X}^\top \mathbf{X} \tilde{c} = \tilde{0} \text{ on the left by } \tilde{c}^\top) \\ &= (\mathbf{X} \tilde{c})^\top (\mathbf{X} \tilde{c}) && \text{(transpose of a product)} \\ &= \|\mathbf{X} \tilde{c}\|^2 && \text{(definition of the Euclidean norm)}\end{aligned}$$

so $\mathbf{X} \tilde{c} = \tilde{0}$. Because $\mathbf{X} \tilde{c} = c_1(\text{column } 1) + \dots + c_p(\text{column } p)$ and the columns of $\mathbf{X}$ are linearly independent, $\tilde{c} = \tilde{0}$. So the only solution of $\mathbf{X}^\top \mathbf{X} \tilde{c} = \tilde{0}$ is $\tilde{c} = \tilde{0}$, which says that the columns of the square matrix $\mathbf{X}^\top \mathbf{X}$ are linearly independent, and a square matrix with linearly independent columns is invertible (Banerjee and Roy 2014, Corollary 5.7, p. 143). $\square$

Example 0.1 (Inverting $\mathbf{X}^\top \mathbf{X}$). For the rank-2 matrix $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ from [Example 2 in Subspaces and Rank](#):

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad (\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix},$$

and multiplying the two out gives $\mathbf{I}_2$.

Definition 0.2 (Hat matrix). For an $n \times p$ design matrix $\mathbf{X}$ (Definition 0.1) with $\text{rank}(\mathbf{X}) = p$, the **hat matrix** is the $n \times n$ matrix

$$\mathbf{H} \stackrel{\text{def}}{=} \underbrace{\mathbf{X}}_{n \times p} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{p \times p} \underbrace{\mathbf{X}^\top}_{p \times n}$$

The inverse exists by Theorem 0.1.

Example 0.2 (The hat matrix of an intercept-only model). With $n = 2$ observations and only an intercept, $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (2×1 , rank 1), so $\mathbf{X}^\top \mathbf{X} = 2$ and

$$\begin{aligned} \mathbf{H} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}. \end{aligned}$$

Then $\mathbf{H}\tilde{\mathbf{y}} = (\bar{y}, \bar{y})$, where $\bar{y} = (y_1 + y_2)/2$: the fitted values of an intercept-only model are the [sample mean](#).

Theorem 0.2 (Hat matrix is a projection matrix). *If $\mathbf{X}$ is an $n \times p$ design matrix with $\text{rank}(\mathbf{X}) = p$, then the hat matrix $\mathbf{H}$ (Definition 0.2) is an orthogonal projection matrix (Definition 8 in [Special Matrices and Decompositions](#)).*

i Proof

Proof. We verify symmetry and idempotency. Both use that $\mathbf{X}^\top \mathbf{X}$ is symmetric,

$$\begin{aligned} (\mathbf{X}^\top \mathbf{X})^\top &= \mathbf{X}^\top (\mathbf{X}^\top)^\top \\ &= \mathbf{X}^\top \mathbf{X} \end{aligned}$$

([Theorem 16 in Matrices](#)), so its inverse is symmetric too ([Corollary 1 in Special Matrices and Decompositions](#)).

Symmetry:

$$\begin{aligned}
\mathbf{H}^\top &= (\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top)^\top && \text{(definition of } \mathbf{H} \text{)} \\
&= (\mathbf{X}^\top)^\top \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transpose of a product, twice)} \\
&= \mathbf{X} \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transposing twice changes nothing)} \\
&= \mathbf{X} \cdot (\mathbf{X}^\top \mathbf{X})^{-1} \cdot \mathbf{X}^\top && \text{(the inverse of a symmetric matrix is symmetric)} \\
&= \mathbf{H} && \text{(definition of } \mathbf{H} \text{)}
\end{aligned}$$

Idempotency:

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \cdot \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of } \mathbf{H} \text{)} \\
&= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(regroup; matrix multiplication is associative)} \\
&= \mathbf{X} \mathbf{I}_p (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of the inverse)} \\
&= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(identity matrix)} \\
&= \mathbf{H} && \text{(definition of } \mathbf{H} \text{)}
\end{aligned}$$

□

Example 0.3 (The intercept-only hat matrix is a projection). For $\mathbf{H} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ from Example 0.2, $\mathbf{H}^\top = \mathbf{H}$, and

$$\begin{aligned}
\mathbf{H}^2 &= \begin{bmatrix} 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \\ 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \end{bmatrix} \\
&= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \\
&= \mathbf{H},
\end{aligned}$$

as Theorem 0.2 says.

Remark 0.2 (Why it is called the hat matrix). The hat matrix gives the fitted values in linear regression:

$$\begin{aligned}
\hat{\tilde{y}} &= \mathbf{X} \hat{\tilde{\beta}} \\
&= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{y} \\
&= \mathbf{H} \tilde{y}.
\end{aligned}$$

Multiplying by $\mathbf{H}$ “puts a hat” on $\tilde{y}$, which is where the name comes from. In Example 0.2, $\mathbf{H}$ puts a hat on $\tilde{y}$ by replacing each y_i with the sample mean $\bar{y}$.

Orthogonal projection onto a subspace

Source

This section is adapted from the second half of Zhou (2024c), used under the MIT License (see the license text in [Section 2 in Subspaces and Rank](#)). It connects the orthogonal projection matrices of [Definition 8 in Special Matrices and Decompositions](#), defined by an algebraic property, to the geometric decomposition of [Theorem 31 in Subspaces and Rank](#).

Definition 0.3 (Orthogonal projection onto a subspace). Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ and $\tilde{y} \in \mathbb{R}^p$. By [Theorem 31 in Subspaces and Rank](#), $\tilde{y} = \tilde{u} + \tilde{v}$ for exactly one $\tilde{u} \in \mathcal{S}$ and $\tilde{v} \in \mathcal{S}^\perp$. The vector $\tilde{u}$ is the **orthogonal projection** of $\tilde{y}$ onto $\mathcal{S}$. It is the projection of $\tilde{y}$ onto $\mathcal{S}$ along $\mathcal{S}^\perp$ ([Definition 18 in Subspaces and Rank](#)).

Example 0.4 (Projecting onto a line in $\mathbb{R}^3$). In [Example 56 in Subspaces and Rank](#), $\tilde{y} = (3, 1, 2)$ splits as $(2, 2, 0) + (1, -1, 2)$, with $(2, 2, 0) \in \mathcal{S} = \text{span}\{(1, 1, 0)\}$ and $(1, -1, 2) \in \mathcal{S}^\perp$. So the orthogonal projection of $(3, 1, 2)$ onto $\mathcal{S}$ is $(2, 2, 0)$. A vector already in $\mathcal{S}$, such as $(5, 5, 0)$, is its own projection, since $(5, 5, 0) = (5, 5, 0) + \tilde{0}$ and $\tilde{0} \in \mathcal{S}^\perp$ ([Theorem 29 in Subspaces and Rank](#), [Theorem 1 in Subspaces and Rank](#)); a vector of $\mathcal{S}^\perp$, such as $(1, -1, 2)$, projects to $\tilde{0}$, since $(1, -1, 2) = \tilde{0} + (1, -1, 2)$ and $\tilde{0} \in \mathcal{S}$ ([Theorem 1 in Subspaces and Rank](#)).

Example 0.5 (A split that is not the orthogonal one). $(3, 1, 2) = (3, 3, 0) + (0, -2, 2)$ also writes $(3, 1, 2)$ as a vector of $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ plus a remainder, but $(1, 1, 0) \cdot (0, -2, 2) = -2 \neq 0$, so the remainder is not in $\mathcal{S}^\perp$, and $(3, 3, 0)$ is not the orthogonal projection of $(3, 1, 2)$ onto $\mathcal{S}$.

Theorem 0.3 (The orthogonal projection is the closest point). *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$, let $\tilde{y} \in \mathbb{R}^p$, and let $\tilde{u}$ be the orthogonal projection of $\tilde{y}$ onto $\mathcal{S}$ ([Definition 0.3](#)). Then for every $\tilde{w} \in \mathcal{S}$,*

$$\|\tilde{y} - \tilde{u}\| \leq \|\tilde{y} - \tilde{w}\|,$$

with equality only when $\tilde{w} = \tilde{u}$.

i Proof

Proof. By Definition 0.3, $\tilde{y} - \tilde{u} \in \mathcal{S}^\perp$. Both $\tilde{u}$ and $\tilde{w}$ are in $\mathcal{S}$, so $\tilde{u} - \tilde{w} = \tilde{u} + (-1)\tilde{w} \in \mathcal{S}$ (Definition 4 in Subspaces and Rank), and therefore $(\tilde{y} - \tilde{u}) \perp (\tilde{u} - \tilde{w})$ (Definition 19 in Subspaces and Rank). Then

$$\begin{aligned}\|\tilde{y} - \tilde{w}\|^2 &= \|(\tilde{y} - \tilde{u}) + (\tilde{u} - \tilde{w})\|^2 \quad (\text{add and subtract } \tilde{u}) \\ &= \|\tilde{y} - \tilde{u}\|^2 + \|\tilde{u} - \tilde{w}\|^2 \quad (\text{Pythagorean theorem, Theorem 2 in Inner Products and Orthogonality}) \\ &\geq \|\tilde{y} - \tilde{u}\|^2, \quad (\|\tilde{u} - \tilde{w}\|^2 \geq 0, \text{Theorem 1 in Inner Products and Orthogonality})\end{aligned}$$

with equality exactly when $\|\tilde{u} - \tilde{w}\| = 0$, that is, when $\tilde{w} = \tilde{u}$ (Theorem 1 in Inner Products and Orthogonality, part 1). Taking nonnegative square roots preserves the inequality. $\square$

Example 0.6 (No point of the line is closer). In Example 0.4, $\tilde{y} = (3, 1, 2)$ has projection $\tilde{u} = (2, 2, 0)$ onto $\mathcal{S} = \text{span}\{(1, 1, 0)\}$, at distance $\|(1, -1, 2)\| = \sqrt{6}$. Another point of $\mathcal{S}$, such as $\tilde{w} = (3, 3, 0)$, is farther:

$$\begin{aligned}\|\tilde{y} - \tilde{w}\| &= \|(0, -2, 2)\| \\ &= \sqrt{8} \\ &> \sqrt{6},\end{aligned}$$

and the difference of squares is

$$\begin{aligned}\|\tilde{u} - \tilde{w}\|^2 &= \|(-1, -1, 0)\|^2 \\ &= 2 \\ &= 8 - 6,\end{aligned}$$

by the Pythagorean theorem (Theorem 2 in Inner Products and Orthogonality), since $\tilde{y} - \tilde{u} \in \mathcal{S}^\perp$ and $\tilde{u} - \tilde{w} \in \mathcal{S}$.

Theorem 0.4 (Projecting with an orthonormal basis). *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ with $\dim(\mathcal{S}) = r \geq 1$, and let $\mathbf{Q}$ be the $p \times r$ matrix whose columns $\tilde{q}_1, \dots, \tilde{q}_r$ are an orthonormal basis of $\mathcal{S}$ (Definition 9 in Inner Products and Orthogonality, Corollary 1 in Inner Products and Orthogonality). Then for every $\tilde{y} \in \mathbb{R}^p$, $\mathbf{Q}\mathbf{Q}^\top \tilde{y}$ is the orthogonal projection of $\tilde{y}$ onto $\mathcal{S}$, and $\mathbf{Q}\mathbf{Q}^\top$ is an orthogonal projection matrix (Definition 8 in Special Matrices and Decompositions).*

i Proof

Proof. $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$. Write q_{ki} for entry k of $\tilde{q}_i$, which is entry (k, i) of $\mathbf{Q}$. Then

$$\begin{aligned} (\mathbf{Q}^\top \mathbf{Q})_{ij} &= \sum_{k=1}^p (\mathbf{Q}^\top)_{ik} q_{kj} \quad (\text{Definition 7 in Matrices}) \\ &= \sum_{k=1}^p q_{ki} q_{kj} \quad (\text{Definition 3 in Matrices}) \\ &= \tilde{q}_i \cdot \tilde{q}_j, \quad (\text{Definition 7 in Vectors}) \end{aligned}$$

which is 1 if $i = j$ and 0 otherwise (Definition 17 in Vectors), so $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$ (Definition 10 in Matrices).

The split. Let $\tilde{u} = \mathbf{Q}\mathbf{Q}^\top \tilde{y}$ and $\tilde{v} = \tilde{y} - \tilde{u}$, so $\tilde{y} = \tilde{u} + \tilde{v}$. $\tilde{u} = \mathbf{Q}(\mathbf{Q}^\top \tilde{y})$ (Theorem 5 in Matrices) is in $\mathcal{C}(\mathbf{Q})$ (Definition 9 in Subspaces and Rank), which is $\text{span}\{\tilde{q}_1, \dots, \tilde{q}_r\} = \mathcal{S}$ (Theorem 12 in Subspaces and Rank). For $\tilde{v}$,

$$\begin{aligned} \mathbf{Q}^\top \tilde{v} &= \mathbf{Q}^\top (\tilde{y} - \mathbf{Q}\mathbf{Q}^\top \tilde{y}) \quad (\text{substitute}) \\ &= \mathbf{Q}^\top \tilde{y} - \mathbf{Q}^\top (\mathbf{Q}\mathbf{Q}^\top \tilde{y}) \quad (\text{Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \mathbf{Q}^\top \tilde{y} - (\mathbf{Q}^\top \mathbf{Q}) \mathbf{Q}^\top \tilde{y} \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{Q}^\top \tilde{y} - \mathbf{I}_r \mathbf{Q}^\top \tilde{y} \quad (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r, \text{ from the first part of the proof}) \\ &= \mathbf{Q}^\top \tilde{y} - \mathbf{Q}^\top \tilde{y} \quad (\text{Theorem 7 in Matrices}) \\ &= \tilde{0}_r, \quad (\text{arithmetic}) \end{aligned}$$

so $\tilde{v} \in \mathcal{N}(\mathbf{Q}^\top)$, and

$$\begin{aligned} \mathcal{N}(\mathbf{Q}^\top) &= \mathcal{C}(\mathbf{Q})^\perp \\ &= \mathcal{S}^\perp \end{aligned}$$

(Theorem 30 in Subspaces and Rank). By the uniqueness in Theorem 31 in Subspaces and Rank, $\tilde{u}$ is the orthogonal projection of $\tilde{y}$ onto $\mathcal{S}$.

$\mathbf{Q}\mathbf{Q}^\top$ is symmetric and idempotent.

$$\begin{aligned} (\mathbf{Q}\mathbf{Q}^\top)^\top &= (\mathbf{Q}^\top)^\top \mathbf{Q}^\top \quad (\text{Theorem 16 in Matrices}) \\ &= \mathbf{Q}\mathbf{Q}^\top, \quad (\text{transposing twice, Definition 3 in Matrices}) \end{aligned}$$

and

$$\begin{aligned} (\mathbf{Q}\mathbf{Q}^\top)(\mathbf{Q}\mathbf{Q}^\top) &= \mathbf{Q}(\mathbf{Q}^\top \mathbf{Q}) \mathbf{Q}^\top \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{Q}\mathbf{I}_r \mathbf{Q}^\top \quad (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r, \text{ from the first part of the proof}) \\ &= \mathbf{Q}\mathbf{Q}^\top. \quad (\text{Theorem 7 in Matrices}) \end{aligned}$$

□

Example 0.7 (The projection matrix onto a line). For $\mathcal{S} = \text{span}\{(1, 1, 0)\}$, the single vector $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ is an orthonormal basis: it spans $\mathcal{S}$, it is nonzero and so linearly independent, and

$$\begin{aligned}\|\tilde{q}_1\| &= \sqrt{\frac{1}{2} + \frac{1}{2} + 0} \\ &= 1.\end{aligned}$$

With $\mathbf{Q} = [\tilde{q}_1]$,

$$\begin{aligned}\mathbf{Q}\mathbf{Q}^\top &= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}.\end{aligned}$$

Applied to $\tilde{y} = (3, 1, 2)$, it gives $(\frac{3+1}{2}, \frac{3+1}{2}, 0) = (2, 2, 0)$, the projection found in Example 0.4.

Theorem 0.5 (A projection matrix projects onto its column space, and is the only one that does).

1. If $\mathbf{P}$ is a $p \times p$ orthogonal projection matrix ([Definition 8 in Special Matrices and Decompositions](#)), then for every $\tilde{y} \in \mathbb{R}^p$, $\mathbf{P}\tilde{y}$ is the orthogonal projection of $\tilde{y}$ onto $\mathcal{C}(\mathbf{P})$ ([Definition 0.3](#)).
2. If $\mathbf{P}_1$ and $\mathbf{P}_2$ are $p \times p$ matrices such that, for every $\tilde{y}$, both $\mathbf{P}_1\tilde{y}$ and $\mathbf{P}_2\tilde{y}$ are the orthogonal projection of $\tilde{y}$ onto the same subspace $\mathcal{S}$, then $\mathbf{P}_1 = \mathbf{P}_2$.

Proof

Proof. Part 1. Write $\tilde{y} = \mathbf{P}\tilde{y} + (\tilde{y} - \mathbf{P}\tilde{y})$. The first term is in $\mathcal{C}(\mathbf{P})$ ([Definition 9 in Subspaces and Rank](#)). For the second,

$$\begin{aligned}\mathbf{P}(\tilde{y} - \mathbf{P}\tilde{y}) &= \mathbf{P}\tilde{y} - \mathbf{P}(\mathbf{P}\tilde{y}) \quad (\text{Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \mathbf{P}\tilde{y} - (\mathbf{P}\mathbf{P})\tilde{y} \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{P}\tilde{y} - \mathbf{P}^2\tilde{y} \quad (\text{Definition 2 in Special Matrices and Decompositions}) \\ &= \mathbf{P}\tilde{y} - \mathbf{P}\tilde{y} \quad (\mathbf{P} \text{ is idempotent}) \\ &= \tilde{0}_p, \quad (\text{arithmetic})\end{aligned}$$

so $\tilde{y} - \mathbf{P}\tilde{y} \in \mathcal{N}(\mathbf{P})$. Since $\mathbf{P} = \mathbf{P}^\top$,

$$\begin{aligned}\mathcal{N}(\mathbf{P}) &= \mathcal{N}(\mathbf{P}^\top) \\ &= \mathcal{C}(\mathbf{P})^\perp\end{aligned}$$

(Theorem 30 in Subspaces and Rank). By the uniqueness in Theorem 31 in Subspaces and Rank, $\mathbf{P}\tilde{\mathbf{y}}$ is the orthogonal projection of $\tilde{\mathbf{y}}$ onto $\mathcal{C}(\mathbf{P})$.

Part 2. The orthogonal projection of a vector onto $\mathcal{S}$ is unique (Definition 0.3), so $\mathbf{P}_1\tilde{\mathbf{e}}_j = \mathbf{P}_2\tilde{\mathbf{e}}_j$ for each indicator vector $\tilde{\mathbf{e}}_j$ (Definition 12 in Vectors). $\mathbf{P}_1\tilde{\mathbf{e}}_j$ is column j of $\mathbf{P}_1$: by Theorem 3 in Subspaces and Rank it is the combination of the columns of $\mathbf{P}_1$ with coefficients the entries of $\tilde{\mathbf{e}}_j$, which are 1 in place j and 0 elsewhere (Definition 12 in Vectors). Likewise for $\mathbf{P}_2$, so the two matrices have the same columns. $\square$

Example 0.8 (Two routes to the same projection matrix). The matrix $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ of Example 10 in Special Matrices and Decompositions is an orthogonal projection matrix, and its column space is $\text{span}\{(1, 0)\}$, the horizontal axis. By part 1, $\mathbf{P}(v_1, v_2) = (v_1, 0)$ is the orthogonal projection onto that axis: the remainder $(0, v_2)$ is orthogonal to $(1, 0)$. The vector $\tilde{\mathbf{q}}_1 = (1, 0)$ has length 1 and spans the axis, so it is an orthonormal basis of it, and with $\mathbf{B} = [\tilde{\mathbf{q}}_1]$,

$$\begin{aligned}\mathbf{B}\mathbf{B}^\top &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \\ &= \mathbf{P},\end{aligned}$$

as part 2 and Theorem 0.4 require.

The oblique projection $\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ of Example 10 in Special Matrices and Decompositions also maps every vector into the horizontal axis. It sends $(0, 1)$ to $(1, 0)$, but the remainder $(0, 1) - (1, 0) = (-1, 1)$ is not orthogonal to the axis, so $(1, 0)$ is not the orthogonal projection of $(0, 1)$ onto the axis, which is $\mathbf{P}(0, 1) = (0, 0)$.

Theorem 0.6 (The hat matrix projects onto the column space of the design matrix). *If $\mathbf{X}$ is an $n \times p$ design matrix with $\text{rank}(\mathbf{X}) = p$, then for every $\tilde{\mathbf{y}} \in \mathbb{R}^n$, $\mathbf{H}\tilde{\mathbf{y}}$ (Definition 0.2) is the orthogonal projection of $\tilde{\mathbf{y}}$ onto $\mathcal{C}(\mathbf{X})$. So $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$ for any $n \times p$ matrix $\mathbf{Q}$ whose columns are an orthonormal basis of $\mathcal{C}(\mathbf{X})$.*

i Proof

Proof. $\mathbf{H}$ is an orthogonal projection matrix (Theorem 0.2), so $\mathbf{H}\tilde{\mathbf{y}}$ is the orthogonal projection of $\tilde{\mathbf{y}}$ onto $\mathcal{C}(\mathbf{H})$ (Theorem 0.5, part 1). It remains to show $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$. $\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$. $\mathbf{H}\tilde{\mathbf{z}} = \mathbf{X}((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\tilde{\mathbf{z}})$ (Theorem 5 in Matrices), which is in $\mathcal{C}(\mathbf{X})$

(Definition 9 in Subspaces and Rank).
 $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$. First,

$$\begin{aligned}
\mathbf{H}\mathbf{X} &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} && \text{(Definition 0.2)} \\
&= \mathbf{X} ((\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X})) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{X} \mathbf{I}_p && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{X}. && \text{(Theorem 7 in Matrices)}
\end{aligned}$$

So for every $\tilde{b} \in \mathbb{R}^p$, $\mathbf{X}\tilde{b} = (\mathbf{H}\mathbf{X})\tilde{b}$, since $\mathbf{H}\mathbf{X} = \mathbf{X}$, and $(\mathbf{H}\mathbf{X})\tilde{b} = \mathbf{H}(\mathbf{X}\tilde{b})$ (Theorem 5 in Matrices), which is in $\mathcal{C}(\mathbf{H})$.

$\mathcal{C}(\mathbf{X})$ has dimension $p \geq 1$ (Theorem 16 in Subspaces and Rank), so an orthonormal basis of it has p vectors (Definition 8 in Subspaces and Rank), and $\mathbf{Q}\mathbf{Q}^\top$ also gives the orthogonal projection onto $\mathcal{C}(\mathbf{X})$ (Theorem 0.4, with n in place of p and $r = p$). By Theorem 0.5, part 2, $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$. $\square$

Example 0.9 (The intercept-only hat matrix, from an orthonormal basis). For the intercept-only design $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ of Example 0.2, $\mathcal{C}(\mathbf{X}) = \text{span}\{(1, 1)\}$, with orthonormal basis $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$, since

$$\begin{aligned}
\|\tilde{q}_1\| &= \sqrt{\frac{1}{2} + \frac{1}{2}} \\
&= 1.
\end{aligned}$$

With $\mathbf{Q} = [\tilde{q}_1]$,

$$\begin{aligned}
\mathbf{Q}\mathbf{Q}^\top &= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix},
\end{aligned}$$

the hat matrix found in Example 0.2. The fitted values $(\bar{y}, \bar{y})$ are the closest point to $\tilde{y}$ on the line of constant vectors (Theorem 0.3), and the residuals $(y_1 - \bar{y}, y_2 - \bar{y})$ are orthogonal to $(1, 1)$:

$$\begin{aligned}
(1, 1) \cdot (y_1 - \bar{y}, y_2 - \bar{y}) &= y_1 + y_2 - 2\bar{y} \\
&= 0,
\end{aligned}$$

since $2\bar{y} = y_1 + y_2$.

Generalized inverses and linear systems

i Source

This section is adapted from Zhou (2024b), used under the MIT License (see the license text in [Section 2 in Subspaces and Rank](#)). The source defers the existence of the Moore-Penrose inverse to the singular value decomposition; the proof here builds it from a rank factorization ([Theorem 23 in Subspaces and Rank](#)) instead, and the section adds the projection $\mathbf{XGX}^\top$ for any generalized inverse $\mathbf{G}$ of $\mathbf{X}^\top \mathbf{X}$, which the source does not cover.

Theorem 0.7 (A scalar factor passes through a matrix product). *For an $m \times k$ matrix $\mathbf{A}$, a $k \times n$ matrix $\mathbf{B}$ and a number c ,*

$$\begin{aligned}\mathbf{A}(c\mathbf{B}) &= c(\mathbf{AB}) \\ &= (c\mathbf{A})\mathbf{B}.\end{aligned}$$

In particular, $\mathbf{A}(\mathbf{B} - \mathbf{C}) = \mathbf{AB} - \mathbf{AC}$ and $(\mathbf{B} - \mathbf{C})\mathbf{D} = \mathbf{BD} - \mathbf{CD}$ for matrices of compatible sizes, where $\mathbf{B} - \mathbf{C} = \mathbf{B} + (-1)\mathbf{C}$ ([Theorem 4 in Matrices](#), [Definition 6 in Matrices](#)).

i Proof

Proof. $\mathbf{A}(c\mathbf{B}) = c(\mathbf{AB})$. For each entry (i, j) ,

$$\begin{aligned}(\mathbf{A}(c\mathbf{B}))_{ij} &= \sum_{l=1}^k a_{il} (c\mathbf{B})_{lj} \quad (\text{Definition 7 in Matrices}) \\ &= \sum_{l=1}^k a_{il} (c b_{lj}) \quad (\text{Definition 6 in Matrices}) \\ &= \sum_{l=1}^k c (a_{il} b_{lj}) \quad (\text{reorder the product of numbers}) \\ &= c \sum_{l=1}^k a_{il} b_{lj} \quad (\text{factor } c \text{ out of the sum}) \\ &= c(\mathbf{AB})_{ij} \quad (\text{Definition 7 in Matrices}) \\ &= (c(\mathbf{AB}))_{ij}. \quad (\text{Definition 6 in Matrices})\end{aligned}$$

$(c\mathbf{A})\mathbf{B} = c(\mathbf{AB})$. For each entry (i, j) ,

$$\begin{aligned}
((c \mathbf{A}) \mathbf{B})_{ij} &= \sum_{l=1}^k (c a_{il}) b_{lj} \quad (\text{Definition 7 in Matrices, Definition 6 in Matrices}) \\
&= c \sum_{l=1}^k a_{il} b_{lj} \quad (\text{factor } c \text{ out of the sum}) \\
&= (c (\mathbf{AB}))_{ij}. \quad (\text{Definition 7 in Matrices, Definition 6 in Matrices})
\end{aligned}$$

Differences.

$$\begin{aligned}
\mathbf{A} (\mathbf{B} - \mathbf{C}) &= \mathbf{A} (\mathbf{B} + (-1) \mathbf{C}) \quad (\text{the difference as a sum}) \\
&= \mathbf{AB} + \mathbf{A} ((-1) \mathbf{C}) \quad (\text{Theorem 6 in Matrices}) \\
&= \mathbf{AB} + (-1) (\mathbf{AC}) \quad (\text{first part}) \\
&= \mathbf{AB} - \mathbf{AC}, \quad (\text{the difference as a sum})
\end{aligned}$$

and $(\mathbf{B} - \mathbf{C}) \mathbf{D} = \mathbf{BD} - \mathbf{CD}$ the same way, using the second law in [Theorem 6 in Matrices](#) and the second part. A column vector is a matrix with one column, so all of this applies with $\mathbf{B}$, $\mathbf{C}$ or $\mathbf{D}$ a vector. $\square$

Example 0.10 (Pulling a scalar out of a product). With $\mathbf{A} = \begin{bmatrix} 1 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$:

$$\begin{aligned}
\mathbf{A} (2 \mathbf{B}) &= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix} \\
&= 22,
\end{aligned}$$

and

$$\begin{aligned}
2 (\mathbf{AB}) &= 2 (3 + 8) \\
&= 22.
\end{aligned}$$

Definition 0.4 (Generalized inverse). A **generalized inverse** of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ such that

$$\underbrace{\mathbf{A}}_{m \times n} \underbrace{\mathbf{G}}_{n \times m} \underbrace{\mathbf{A}}_{m \times n} = \mathbf{A}.$$

Example 0.11 (A matrix with many generalized inverses). Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{G} =$

$$\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}. \text{ Then}$$

$$\begin{aligned}\mathbf{A}\mathbf{G}\mathbf{A} &= \begin{bmatrix} g_{11} & g_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad (\text{Definition 7 in Matrices, for } \mathbf{A}\mathbf{G}) \\ &= \begin{bmatrix} g_{11} & 0 \\ 0 & 0 \end{bmatrix}, \quad (\text{Definition 7 in Matrices})\end{aligned}$$

which equals $\mathbf{A}$ exactly when $g_{11} = 1$. So every matrix $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ is a generalized inverse of $\mathbf{A}$: a generalized inverse need not be unique. The zero matrix is not one, since its $(1, 1)$ entry is 0.

Theorem 0.8 (An invertible matrix has only one generalized inverse). *If $\mathbf{A}$ is an invertible $p \times p$ matrix (Definition 6 in Special Matrices and Decompositions), its only generalized inverse is $\mathbf{A}^{-1}$.*

i Proof

Proof.

$$\begin{aligned}\mathbf{A}\mathbf{A}^{-1}\mathbf{A} &= \mathbf{I}_p\mathbf{A} \\ &= \mathbf{A}\end{aligned}$$

(Definition 5 in Special Matrices and Decompositions, Theorem 7 in Matrices), so $\mathbf{A}^{-1}$ is a generalized inverse. If $\mathbf{G}$ is any generalized inverse, then

$$\begin{aligned}\mathbf{G} &= \mathbf{I}_p\mathbf{G}\mathbf{I}_p && (\text{Theorem 7 in Matrices}) \\ &= (\mathbf{A}^{-1}\mathbf{A})\mathbf{G}(\mathbf{A}\mathbf{A}^{-1}) && (\text{Definition 5 in Special Matrices and Decompositions}) \\ &= \mathbf{A}^{-1}(\mathbf{A}\mathbf{G}\mathbf{A})\mathbf{A}^{-1} && (\text{Theorem 5 in Matrices}) \\ &= \mathbf{A}^{-1}\mathbf{A}\mathbf{A}^{-1} && (\text{Definition 0.4}) \\ &= \mathbf{I}_p\mathbf{A}^{-1} && (\text{Definition 5 in Special Matrices and Decompositions}) \\ &= \mathbf{A}^{-1}. && (\text{Theorem 7 in Matrices})\end{aligned}$$

□

Example 0.12 (The generalized inverse of an invertible matrix). For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ of Example 5 in Special Matrices and Decompositions, the only generalized inverse is $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$. The matrix of Example 0.11, by contrast, has infinitely many, so by Theorem 0.8 it is singular.

Theorem 0.9 (A generalized inverse solves every solvable system). *Let $\mathbf{A}$ be $m \times n$, let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$ (Definition 0.4), and let $\tilde{\mathbf{b}} \in \mathbb{R}^m$. If $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ has a solution, then $\mathbf{G}\tilde{\mathbf{b}}$ is a solution.*

Proof

Proof. Let $\tilde{\mathbf{x}}_0$ be a solution, so $\mathbf{A}\tilde{\mathbf{x}}_0 = \tilde{\mathbf{b}}$. Then

$$\begin{aligned} \mathbf{A}(\mathbf{G}\tilde{\mathbf{b}}) &= \mathbf{A}\mathbf{G}(\mathbf{A}\tilde{\mathbf{x}}_0) && \text{(substitute } \tilde{\mathbf{b}} = \mathbf{A}\tilde{\mathbf{x}}_0) \\ &= (\mathbf{A}\mathbf{G}\mathbf{A})\tilde{\mathbf{x}}_0 && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{A}\tilde{\mathbf{x}}_0 && \text{(Definition 0.4)} \\ &= \tilde{\mathbf{b}}. && (\tilde{\mathbf{x}}_0 \text{ is a solution}) \end{aligned}$$

□

Example 0.13 (Solving with a generalized inverse). With $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and its generalized inverse $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$ (Example 0.11), the system $\mathbf{A}\tilde{\mathbf{x}} = (2, 0)$ has the solution $(2, 0)$, and $\mathbf{G}(2, 0) = (2, 14)$ is a solution too: $\mathbf{A}(2, 14) = (2, 0)$. The system $\mathbf{A}\tilde{\mathbf{x}} = (2, 1)$ has no solution, since the second entry of $\mathbf{A}\tilde{\mathbf{x}}$ is always 0; there $\mathbf{G}(2, 1) = (7, 14)$ gives $\mathbf{A}(7, 14) = (7, 0) \neq (2, 1)$.

Definition 0.5 (Moore-Penrose inverse). A **Moore-Penrose inverse** of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ satisfying all four conditions

1. $\mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A}$;
2. $\mathbf{G}\mathbf{A}\mathbf{G} = \mathbf{G}$;
3. $(\mathbf{A}\mathbf{G})^\top = \mathbf{A}\mathbf{G}$;
4. $(\mathbf{G}\mathbf{A})^\top = \mathbf{G}\mathbf{A}$.

Example 0.14 (Checking the four conditions). For $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, the matrix $\mathbf{G} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ satisfies all four:

$$\begin{aligned} \mathbf{A}\mathbf{G} &= \mathbf{G}\mathbf{A} \\ &= \mathbf{A} \end{aligned}$$

and $\mathbf{A}^2 = \mathbf{A}$, so conditions 1 and 2 read $\mathbf{A} = \mathbf{A}$, and conditions 3 and 4 hold because $\mathbf{A}$ is symmetric.

The generalized inverse $\mathbf{G}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ of $\mathbf{A}$ (Example 0.11) is not a Moore-Penrose inverse: $\mathbf{A}\mathbf{G}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, which is not symmetric, so condition 3 fails.

Theorem 0.10 (A Moore-Penrose inverse is a generalized inverse). *If $\mathbf{G}$ is a Moore-Penrose inverse of an $m \times n$ matrix $\mathbf{A}$ (Definition 0.5), then $\mathbf{G}$ is a generalized inverse of $\mathbf{A}$ (Definition 0.4).*

i Proof

Proof. A generalized inverse of $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ with $\mathbf{AGA} = \mathbf{A}$. A Moore-Penrose inverse is an $n \times m$ matrix, and condition 1 of Definition 0.5 is $\mathbf{AGA} = \mathbf{A}$. $\square$

Example 0.15 (The Moore-Penrose inverse of a rank-one matrix). The 3×2 matrix $\mathbf{A}$ below has rank 1, because its second column is twice its first, so it has no ordinary inverse. The function `MASS::ginv()` computes its Moore-Penrose inverse $\mathbf{G}$. The code checks the four conditions of Definition 0.5, of which the first is the equation that makes $\mathbf{G}$ a generalized inverse.

```
A <- cbind(c(1, 2, 3), c(2, 4, 6))
G <- MASS::ginv(A)
conditions <- c(
  "1: AGA = A" = isTRUE(all.equal(A %*% G %*% A, A)),
  "2: GAG = G" = isTRUE(all.equal(G %*% A %*% G, G)),
  "3: AG symmetric" = isTRUE(all.equal(A %*% G, t(A %*% G))),
  "4: GA symmetric" = isTRUE(all.equal(G %*% A, t(G %*% A)))
)
conditions
#>      1: AGA = A      2: GAG = G 3: AG symmetric 4: GA symmetric
#>           TRUE           TRUE           TRUE           TRUE
round(G, 4)
#>      [,1] [,2] [,3]
#> [1,] 0.0143 0.0286 0.0429
#> [2,] 0.0286 0.0571 0.0857
```

All 4 of 4 conditions hold, so $\mathbf{G}$ is a Moore-Penrose inverse of $\mathbf{A}$. By Theorem 0.10 it is also a generalized inverse, which is condition 1: $\mathbf{AGA} = \mathbf{A}$ up to rounding error.

Theorem 0.11 (A matrix has at most one Moore-Penrose inverse). *If $\mathbf{G}_1$ and $\mathbf{G}_2$ are both Moore-Penrose inverses of an $m \times n$ matrix $\mathbf{A}$ (Definition 0.5), then $\mathbf{G}_1 = \mathbf{G}_2$.*

i Proof

Proof. $\mathbf{A}\mathbf{G}_1 = \mathbf{A}\mathbf{G}_2$.

$$\begin{aligned}
 \mathbf{A}\mathbf{G}_1 &= (\mathbf{A}\mathbf{G}_1)^\top && \text{(condition 3 for } \mathbf{G}_1) \\
 &= \mathbf{G}_1^\top \mathbf{A}^\top && \text{(Theorem 16 in Matrices)} \\
 &= \mathbf{G}_1^\top (\mathbf{A}\mathbf{G}_2\mathbf{A})^\top && \text{(condition 1 for } \mathbf{G}_2) \\
 &= \mathbf{G}_1^\top \mathbf{A}^\top \mathbf{G}_2^\top \mathbf{A}^\top && \text{(Theorem 16 in Matrices, twice)} \\
 &= (\mathbf{G}_1^\top \mathbf{A}^\top) (\mathbf{G}_2^\top \mathbf{A}^\top) && \text{(Theorem 5 in Matrices)} \\
 &= (\mathbf{A}\mathbf{G}_1)^\top (\mathbf{A}\mathbf{G}_2)^\top && \text{(Theorem 16 in Matrices, twice)} \\
 &= (\mathbf{A}\mathbf{G}_1)(\mathbf{A}\mathbf{G}_2) && \text{(condition 3 for } \mathbf{G}_1 \text{ and } \mathbf{G}_2) \\
 &= (\mathbf{A}\mathbf{G}_1\mathbf{A})\mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
 &= \mathbf{A}\mathbf{G}_2. && \text{(condition 1 for } \mathbf{G}_1)
 \end{aligned}$$

$\mathbf{G}_1\mathbf{A} = \mathbf{G}_2\mathbf{A}$.

$$\begin{aligned}
 \mathbf{G}_1\mathbf{A} &= (\mathbf{G}_1\mathbf{A})^\top && \text{(condition 4 for } \mathbf{G}_1) \\
 &= \mathbf{A}^\top \mathbf{G}_1^\top && \text{(Theorem 16 in Matrices)} \\
 &= (\mathbf{A}\mathbf{G}_2\mathbf{A})^\top \mathbf{G}_1^\top && \text{(condition 1 for } \mathbf{G}_2) \\
 &= \mathbf{A}^\top \mathbf{G}_2^\top \mathbf{A}^\top \mathbf{G}_1^\top && \text{(Theorem 16 in Matrices, twice)} \\
 &= (\mathbf{A}^\top \mathbf{G}_2^\top) (\mathbf{A}^\top \mathbf{G}_1^\top) && \text{(Theorem 5 in Matrices)} \\
 &= (\mathbf{G}_2\mathbf{A})^\top (\mathbf{G}_1\mathbf{A})^\top && \text{(Theorem 16 in Matrices, twice)} \\
 &= (\mathbf{G}_2\mathbf{A})(\mathbf{G}_1\mathbf{A}) && \text{(condition 4 for } \mathbf{G}_2 \text{ and } \mathbf{G}_1) \\
 &= \mathbf{G}_2(\mathbf{A}\mathbf{G}_1\mathbf{A}) && \text{(Theorem 5 in Matrices)} \\
 &= \mathbf{G}_2\mathbf{A}. && \text{(condition 1 for } \mathbf{G}_1)
 \end{aligned}$$

Conclusion.

$$\begin{aligned}
\mathbf{G}_1 &= \mathbf{G}_1 \mathbf{A} \mathbf{G}_1 && \text{(condition 2 for } \mathbf{G}_1) \\
&= \mathbf{G}_1 (\mathbf{A} \mathbf{G}_1) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_1 (\mathbf{A} \mathbf{G}_2) && \text{(the first step)} \\
&= (\mathbf{G}_1 \mathbf{A}) \mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{G}_2 \mathbf{A}) \mathbf{G}_2 && \text{(the second step)} \\
&= \mathbf{G}_2 \mathbf{A} \mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_2. && \text{(condition 2 for } \mathbf{G}_2)
\end{aligned}$$

□

Example 0.16 (Only one of the many generalized inverses qualifies). The matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ has infinitely many generalized inverses $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ (Example 0.11), and $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ is a Moore-Penrose inverse (Example 0.14). By Theorem 0.11 it is the only one, so every other choice of g_{12}, g_{21}, g_{22} breaks condition 2, 3 or 4. For instance, $g_{21} = 1$ (the rest 0) gives $\mathbf{GA} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, which is not symmetric.

Theorem 0.12 (Every matrix has a Moore-Penrose inverse). *Every $m \times n$ matrix $\mathbf{A}$ has a Moore-Penrose inverse (Definition 0.5), written $\mathbf{A}^+$; it is unique by Theorem 0.11.*

- If $\mathbf{A} = \mathbf{0}_{m \times n}$, then $\mathbf{A}^+ = \mathbf{0}_{n \times m}$.
- If $\text{rank}(\mathbf{A}) = r \geq 1$ and $\mathbf{A} = \mathbf{CR}$ is a rank factorization (Definition 15 in Subspaces and Rank), then

$$\underbrace{\mathbf{A}^+}_{n \times m} = \underbrace{\mathbf{R}^\top}_{n \times r} \underbrace{(\mathbf{R}\mathbf{R}^\top)^{-1}}_{r \times r} \underbrace{(\mathbf{C}^\top \mathbf{C})^{-1}}_{r \times r} \underbrace{\mathbf{C}^\top}_{r \times m}.$$

i Proof

Proof. **The two cases cover every matrix.** If $\mathbf{A} \neq \mathbf{0}_{m \times n}$, it has a nonzero column, which is linearly independent on its own ($c\tilde{v} = \tilde{0}$ with $\tilde{v} \neq \tilde{0}$ forces $c = 0$), so $\text{rank}(\mathbf{A}) \geq 1$ (Definition 2 in Subspaces and Rank).

The zero matrix. With $\mathbf{A} = \mathbf{0}_{m \times n}$ and $\mathbf{G} = \mathbf{0}_{n \times m}$, every product in conditions 1 to 4 is a zero matrix, and a zero matrix equals its own transpose, so all four conditions hold.

The two inverses exist. Now suppose $\text{rank}(\mathbf{A}) = r \geq 1$. A rank factorization exists (Theorem 23 in Subspaces and Rank). $\mathbf{C}$ is $m \times r$ and $\mathbf{R}$ is $r \times n$. Both have rank r :

$$\begin{aligned}
r &= \text{rank}(\mathbf{C}\mathbf{R}) \quad (\mathbf{A} = \mathbf{C}\mathbf{R}) \\
&\leq \text{rank}(\mathbf{C}) \quad (\text{Theorem 18 in Subspaces and Rank}) \\
&\leq r, \quad (\text{Corollary 1 in Subspaces and Rank, } \mathbf{C} \text{ has } r \text{ columns})
\end{aligned}$$

and, using $(\mathbf{C}\mathbf{R})^\top = \mathbf{R}^\top \mathbf{C}^\top$ (Theorem 16 in Matrices),

$$\begin{aligned}
r &= \text{rank}(\mathbf{R}^\top \mathbf{C}^\top) \quad (\text{Theorem 19 in Subspaces and Rank}) \\
&\leq \text{rank}(\mathbf{R}^\top) \quad (\text{Theorem 18 in Subspaces and Rank}) \\
&\leq r. \quad (\text{Corollary 1 in Subspaces and Rank, } \mathbf{R}^\top \text{ has } r \text{ columns})
\end{aligned}$$

So $\mathbf{C}$ and $\mathbf{R}^\top$ have full column rank r , and $\mathbf{C}^\top \mathbf{C}$ and $\mathbf{R}\mathbf{R}^\top = (\mathbf{R}^\top)^\top \mathbf{R}^\top$ are invertible (Theorem 0.1). They are symmetric:

$$\begin{aligned}
(\mathbf{C}^\top \mathbf{C})^\top &= \mathbf{C}^\top (\mathbf{C}^\top)^\top \\
&= \mathbf{C}^\top \mathbf{C}
\end{aligned}$$

(Theorem 16 in Matrices, Definition 3 in Matrices), and likewise for $\mathbf{R}\mathbf{R}^\top$. Write $\mathbf{S} \stackrel{\text{def}}{=} (\mathbf{R}\mathbf{R}^\top)^{-1}$ and $\mathbf{T} \stackrel{\text{def}}{=} (\mathbf{C}^\top \mathbf{C})^{-1}$, both symmetric (Corollary 1 in Special Matrices and Decompositions), and $\mathbf{G} \stackrel{\text{def}}{=} \mathbf{R}^\top \mathbf{S} \mathbf{T} \mathbf{C}^\top$.

Two products.

$$\begin{aligned}
\mathbf{A}\mathbf{G} &= \mathbf{C}\mathbf{R}\mathbf{R}^\top \mathbf{S} \mathbf{T} \mathbf{C}^\top \quad (\text{substitute } \mathbf{A} \text{ and } \mathbf{G}) \\
&= \mathbf{C}(\mathbf{R}\mathbf{R}^\top) \mathbf{S} \mathbf{T} \mathbf{C}^\top \quad (\text{Theorem 5 in Matrices}) \\
&= \mathbf{C}\mathbf{I}_r \mathbf{T} \mathbf{C}^\top \quad (\text{Definition 5 in Special Matrices and Decompositions}) \\
&= \mathbf{C} \mathbf{T} \mathbf{C}^\top, \quad (\text{Theorem 7 in Matrices})
\end{aligned}$$

$$\begin{aligned}
\mathbf{G}\mathbf{A} &= \mathbf{R}^\top \mathbf{S} \mathbf{T} \mathbf{C}^\top \mathbf{C} \mathbf{R} \quad (\text{substitute } \mathbf{G} \text{ and } \mathbf{A}) \\
&= \mathbf{R}^\top \mathbf{S} \mathbf{T} (\mathbf{C}^\top \mathbf{C}) \mathbf{R} \quad (\text{Theorem 5 in Matrices}) \\
&= \mathbf{R}^\top \mathbf{S} \mathbf{I}_r \mathbf{R} \quad (\text{Definition 5 in Special Matrices and Decompositions}) \\
&= \mathbf{R}^\top \mathbf{S} \mathbf{R}. \quad (\text{Theorem 7 in Matrices})
\end{aligned}$$

Conditions 3 and 4.

$$\begin{aligned}
(\mathbf{A}\mathbf{G})^\top &= (\mathbf{C} \mathbf{T} \mathbf{C}^\top)^\top \quad (\text{first product}) \\
&= (\mathbf{C} (\mathbf{T} \mathbf{C}^\top))^\top \quad (\text{Theorem 5 in Matrices}) \\
&= (\mathbf{T} \mathbf{C}^\top)^\top \mathbf{C}^\top \quad (\text{Theorem 16 in Matrices}) \\
&= (\mathbf{C}^\top)^\top \mathbf{T}^\top \mathbf{C}^\top \quad (\text{Theorem 16 in Matrices}) \\
&= \mathbf{C} \mathbf{T}^\top \mathbf{C}^\top \quad (\text{Definition 3 in Matrices}) \\
&= \mathbf{C} \mathbf{T} \mathbf{C}^\top, \quad (\mathbf{T} \text{ is symmetric})
\end{aligned}$$

$$\begin{aligned}
(\mathbf{GA})^\top &= (\mathbf{R}^\top \mathbf{SR})^\top && \text{(second product)} \\
&= (\mathbf{R}^\top (\mathbf{SR}))^\top && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{SR})^\top (\mathbf{R}^\top)^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{S}^\top (\mathbf{R}^\top)^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{S}^\top \mathbf{R} && \text{(Definition 3 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{SR}. && \text{(\mathbf{S} is symmetric)}
\end{aligned}$$

Condition 1.

$$\begin{aligned}
\mathbf{AGA} &= \mathbf{CTC}^\top \mathbf{CR} && \text{(first product)} \\
&= \mathbf{CT}(\mathbf{C}^\top \mathbf{C}) \mathbf{R} && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{CI}_r \mathbf{R} && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{A}. && \text{(Theorem 7 in Matrices)}
\end{aligned}$$

Condition 2.

$$\begin{aligned}
\mathbf{GAG} &= \mathbf{R}^\top \mathbf{SR} \mathbf{R}^\top \mathbf{STC}^\top && \text{(second product)} \\
&= \mathbf{R}^\top \mathbf{S} (\mathbf{RR}^\top) \mathbf{STC}^\top && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{I}_r \mathbf{STC}^\top && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{G}. && \text{(Theorem 7 in Matrices)}
\end{aligned}$$

□

Example 0.17 (The Moore-Penrose inverse of a rank-one matrix). For $\mathbf{A}$ in [Example 22 in Subspaces and Rank](#), the rank factorization $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$ ([Example 40 in Subspaces and Rank](#)) gives

$$\begin{aligned}
\mathbf{C}^\top \mathbf{C} &= 1 + 9 \\
&= 10
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{RR}^\top &= 1 + 4 + 4 \\
&= 9,
\end{aligned}$$

so

$$\begin{aligned}\mathbf{A}^+ &= \frac{1}{9 \cdot 10} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \begin{bmatrix} 1 & 3 \end{bmatrix} \\ &= \frac{1}{90} \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}.\end{aligned}$$

As a check on condition 1,

$$\begin{aligned}\mathbf{A}\mathbf{A}^+ &= \mathbf{C}\mathbf{T}\mathbf{C}^\top \\ &= \frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix},\end{aligned}$$

and its first row times $\mathbf{A}$ is $\frac{1}{10}(1 \cdot (1, -2, -2) + 3 \cdot (3, -6, -6)) = (1, -2, -2)$, the first row of $\mathbf{A}$; the second row is 3 times the first, like the second row of $\mathbf{A}$.

Definition 0.6 (Consistent linear system). A linear system ([Definition 12 in Matrices](#)) $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$, with $\mathbf{A}$ an $m \times n$ matrix and $\tilde{\mathbf{b}} \in \mathbb{R}^m$, is **consistent** if it has at least one solution $\tilde{\mathbf{x}} \in \mathbb{R}^n$, and **inconsistent** otherwise.

Example 0.18 (A consistent system and an inconsistent one). With $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\mathbf{A}(x_1, x_2) = (x_1, 0)$ ([Definition 11 in Matrices](#)).

- $\mathbf{A}\tilde{\mathbf{x}} = (2, 0)$ is consistent: $\tilde{\mathbf{x}} = (2, 0)$ is a solution.
- $\mathbf{A}\tilde{\mathbf{x}} = (2, 1)$ is inconsistent: the second entry of $\mathbf{A}\tilde{\mathbf{x}}$ is always 0, never 1.

Theorem 0.13 (When a linear system has a solution). Let $\mathbf{A}$ be $m \times n$, let $\tilde{\mathbf{b}} \in \mathbb{R}^m$, and let $\mathbf{G}$ be any generalized inverse of $\mathbf{A}$ ([Definition 0.4](#)). The following statements are equivalent:

1. $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ is consistent ([Definition 0.6](#));
2. $\tilde{\mathbf{b}} \in \mathcal{C}(\mathbf{A})$ ([Definition 9 in Subspaces and Rank](#));
3. $\mathbf{A}\mathbf{G}\tilde{\mathbf{b}} = \tilde{\mathbf{b}}$.

i Proof

Proof. **1 and 2 are equivalent.** Statement 1 says $\tilde{\mathbf{b}} = \mathbf{A}\tilde{\mathbf{x}}$ for some $\tilde{\mathbf{x}}$ ([Definition 0.6](#)), and $\mathcal{C}(\mathbf{A})$ is the set of vectors $\mathbf{A}\tilde{\mathbf{x}}$ ([Definition 9 in Subspaces and Rank](#)), so $\tilde{\mathbf{b}}$ is in it exactly when $\tilde{\mathbf{b}} = \mathbf{A}\tilde{\mathbf{x}}$ for some $\tilde{\mathbf{x}}$.

1 implies 3. By Theorem 0.9, $\mathbf{G}\tilde{\mathbf{b}}$ is a solution, so $\mathbf{A}(\mathbf{G}\tilde{\mathbf{b}}) = \tilde{\mathbf{b}}$, which is statement 3 (Theorem 5 in Matrices).

3 implies 1. If $\mathbf{A}\mathbf{G}\tilde{\mathbf{b}} = \tilde{\mathbf{b}}$, then $\tilde{\mathbf{x}} = \mathbf{G}\tilde{\mathbf{b}}$ is a solution, since $\mathbf{A}(\mathbf{G}\tilde{\mathbf{b}}) = (\mathbf{A}\mathbf{G})\tilde{\mathbf{b}}$ (Theorem 5 in Matrices). $\square$

Example 0.19 (Testing consistency with a generalized inverse). For $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and the generalized inverse $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$, $\mathbf{A}\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 0 & 0 \end{bmatrix}$.

- $\tilde{\mathbf{b}} = (2, 0)$:

$$\begin{aligned} \mathbf{A}\mathbf{G}\tilde{\mathbf{b}} &= (2, 0) \\ &= \tilde{\mathbf{b}}, \end{aligned}$$

so the system is consistent.

- $\tilde{\mathbf{b}} = (2, 1)$: $\mathbf{A}\mathbf{G}\tilde{\mathbf{b}} = (7, 0) \neq \tilde{\mathbf{b}}$, so it is not, as Example 0.13 found directly.

Theorem 0.14 (All solutions of a linear system). *Let $\mathbf{A}$ be $m \times n$ with a generalized inverse $\mathbf{G}$ (Definition 0.4).*

1. $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I}_n - \mathbf{GA})$: $\tilde{\mathbf{x}}$ solves $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{0}}_m$ exactly when $\tilde{\mathbf{x}} = (\mathbf{I}_n - \mathbf{GA})\tilde{\mathbf{q}}$ for some $\tilde{\mathbf{q}} \in \mathbb{R}^n$.
2. If $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ is consistent (Definition 0.6), then $\tilde{\mathbf{x}}$ is a solution exactly when

$$\tilde{\mathbf{x}} = \mathbf{G}\tilde{\mathbf{b}} + (\mathbf{I}_n - \mathbf{GA})\tilde{\mathbf{q}} \quad \text{for some } \tilde{\mathbf{q}} \in \mathbb{R}^n :$$

one particular solution plus a vector of $\mathcal{N}(\mathbf{A})$.

i Proof

Proof. **Part 1.** First,

$$\begin{aligned} \mathbf{A}(\mathbf{I}_n - \mathbf{GA}) &= \mathbf{A}\mathbf{I}_n - \mathbf{A}(\mathbf{GA}) && \text{(Theorem 0.7)} \\ &= \mathbf{A}\mathbf{I}_n - \mathbf{AGA} && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{A} - \mathbf{AGA} && \text{(Theorem 7 in Matrices)} \\ &= \mathbf{A} - \mathbf{A} && \text{(Definition 0.4)} \\ &= \mathbf{0}_{m \times n}, && \text{(arithmetic)} \end{aligned}$$

so $\mathbf{A}(\mathbf{I}_n - \mathbf{GA})\tilde{q} = \tilde{0}_m$ for every $\tilde{q}$ (Theorem 5 in Matrices), and $\mathcal{C}(\mathbf{I}_n - \mathbf{GA}) \subseteq \mathcal{N}(\mathbf{A})$. Conversely, if $\mathbf{A}\tilde{x} = \tilde{0}_m$, then

$$\begin{aligned} (\mathbf{I}_n - \mathbf{GA})\tilde{x} &= \mathbf{I}_n\tilde{x} - (\mathbf{GA})\tilde{x} && \text{(Theorem 0.7, with } \tilde{x} \text{ as the last factor)} \\ &= \tilde{x} - (\mathbf{GA})\tilde{x} && \text{(Theorem 7 in Matrices)} \\ &= \tilde{x} - \mathbf{G}(\mathbf{A}\tilde{x}) && \text{(Theorem 5 in Matrices)} \\ &= \tilde{x} - \mathbf{G}\tilde{0}_m && (\mathbf{A}\tilde{x} = \tilde{0}_m) \\ &= \tilde{x}, && (\mathbf{G}\tilde{0}_m = \tilde{0}_n) \end{aligned}$$

so $\tilde{x}$ is in $\mathcal{C}(\mathbf{I}_n - \mathbf{GA})$, with $\tilde{q} = \tilde{x}$.

Part 2. $\mathbf{G}\tilde{b}$ is a solution (Theorem 0.9). For any $\tilde{x}$,

$$\begin{aligned} \mathbf{A}(\tilde{x} - \mathbf{G}\tilde{b}) &= \mathbf{A}\tilde{x} - \mathbf{A}(\mathbf{G}\tilde{b}) && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \mathbf{A}\tilde{x} - \mathbf{AG}\tilde{b} && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{A}\tilde{x} - \tilde{b}, && \text{(Theorem 0.13, statement 3)} \end{aligned}$$

so $\tilde{x}$ solves $\mathbf{A}\tilde{x} = \tilde{b}$ exactly when $\tilde{x} - \mathbf{G}\tilde{b} \in \mathcal{N}(\mathbf{A})$, that is, by part 1, exactly when $\tilde{x} - \mathbf{G}\tilde{b} = (\mathbf{I}_n - \mathbf{GA})\tilde{q}$ for some $\tilde{q}$. $\square$

Example 0.20 (All solutions of a consistent system). For $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$ and $\tilde{b} = (2, 0)$: $\mathbf{G}\tilde{b} = (2, 14)$, $\mathbf{GA} = \begin{bmatrix} 1 & 0 \\ 7 & 0 \end{bmatrix}$, and $\mathbf{I}_2 - \mathbf{GA} = \begin{bmatrix} 0 & 0 \\ -7 & 1 \end{bmatrix}$, so

$$\tilde{x} = (2, 14) + (0, -7q_1 + q_2), \quad q_1, q_2 \in \mathbb{R}.$$

Since $-7q_1 + q_2$ can be any number, the solutions are exactly the vectors $(2, t)$, $t \in \mathbb{R}$, which is what

$$\begin{aligned} \mathbf{A}(x_1, x_2) &= (x_1, 0) \\ &= (2, 0) \end{aligned}$$

says directly.

Corollary 0.1 (Solvable for every right-hand side, and solvable uniquely). *Let $\mathbf{A}$ be $m \times n$.*

1. $\mathbf{A}\tilde{x} = \tilde{b}$ is consistent for every $\tilde{b} \in \mathbb{R}^m$ exactly when $\text{rank}(\mathbf{A}) = m$.
2. A consistent system $\mathbf{A}\tilde{x} = \tilde{b}$ has exactly one solution exactly when $\text{rank}(\mathbf{A}) = n$.

i Proof

Proof. Part 1. By Theorem 0.13, the system is consistent for every $\tilde{b}$ exactly when $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$. $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$ (Theorem 12 in Subspaces and Rank) with dimension $\text{rank}(\mathbf{A})$ (Theorem 16 in Subspaces and Rank), and $\dim(\mathbb{R}^m) = m$ (Example 16 in Subspaces and Rank). If the dimension is m , then $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$ (Theorem 11 in Subspaces and Rank); if $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$, the dimension is m .

Part 2. By Theorem 0.14, the solutions are a particular solution plus the vectors of $\mathcal{N}(\mathbf{A})$, so there is exactly one solution exactly when $\mathcal{N}(\mathbf{A}) = \{\tilde{0}_n\}$, that is, when $\text{nullity}(\mathbf{A}) = 0$ (a subspace of dimension 0 has the empty list as a basis, whose span is $\{\tilde{0}\}$, by Definition 8 in Subspaces and Rank and Definition 5 in Subspaces and Rank; and $\{\tilde{0}\}$ has dimension 0, by Example 16 in Subspaces and Rank). By Theorem 17 in Subspaces and Rank, that is when $\text{rank}(\mathbf{A}) = n$. $\square$

Example 0.21 (Three systems).

- The 2×3 matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of Example 34 in Subspaces and Rank has rank $2 = m$, so every system with it is consistent, but $2 < 3 = n$, so the solutions are never unique.
- The 3×2 matrix $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ of Example 0.1 has rank $2 = n$, so a consistent system has one solution, but $2 < 3 = m$, so some systems are inconsistent: $(1, 0, 0) \notin \mathcal{C}(\mathbf{X})$, because $a(1, 1, 1) + b(1, 2, 3)$ has equal differences between consecutive entries, and $(1, 0, 0)$ does not.
- An invertible $p \times p$ matrix $\mathbf{A}$ has rank p : if $\mathbf{A}\tilde{x} = \tilde{0}$ then

$$\begin{aligned} \tilde{x} &= \mathbf{A}^{-1}\mathbf{A}\tilde{x} \\ &= \tilde{0}, \end{aligned}$$

so its nullity is 0 and its rank is p (Theorem 17 in Subspaces and Rank). So every system with it is consistent (part 1, with $p = m$) and has exactly one solution (part 2, with $p = n$), namely $\mathbf{A}^{-1}\tilde{b}$, since $\mathbf{A}(\mathbf{A}^{-1}\tilde{b}) = \tilde{b}$.

Theorem 0.15 (The projection onto a column space, with any generalized inverse). *Let $\mathbf{X}$ be an $n \times p$ matrix of any rank, and let $\mathbf{G}$ be any generalized inverse of $\mathbf{X}^\top \mathbf{X}$ (Definition 0.4). Then for every $\tilde{y} \in \mathbb{R}^n$, $\mathbf{X}\mathbf{G}\mathbf{X}^\top \tilde{y}$ is the orthogonal projection of $\tilde{y}$ onto*

$\mathcal{C}(\mathbf{X})$ (Definition 0.3). So the matrix $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ is the same for every choice of $\mathbf{G}$, and when $\text{rank}(\mathbf{X}) = p$ it is the hat matrix $\mathbf{H}$ (Definition 0.2).

i Proof

Proof. $\mathbf{X}^\top \tilde{\mathbf{y}}$ is in $\mathcal{C}(\mathbf{X}^\top \mathbf{X})$. $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X}^\top)$, since $(\mathbf{X}^\top \mathbf{X}) \tilde{\mathbf{z}} = \mathbf{X}^\top (\mathbf{X} \tilde{\mathbf{z}})$ (Theorem 5 in Matrices). Both are subspaces of $\mathbb{R}^p$ (Theorem 12 in Subspaces and Rank) with the same dimension: $\text{rank}(\mathbf{X}^\top \mathbf{X}) = \text{rank}(\mathbf{X}^\top)$ (Theorem 19 in Subspaces and Rank, Theorem 16 in Subspaces and Rank). So $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$ (Theorem 11 in Subspaces and Rank), and since $\mathbf{X}^\top \tilde{\mathbf{y}} \in \mathcal{C}(\mathbf{X}^\top)$ (Definition 9 in Subspaces and Rank), $\mathbf{X}^\top \tilde{\mathbf{y}} = \mathbf{X}^\top \mathbf{X} \tilde{\mathbf{z}}$ for some $\tilde{\mathbf{z}} \in \mathbb{R}^p$.

The split. Let

$$\begin{aligned} \tilde{\mathbf{u}} &= \mathbf{X}\mathbf{G}\mathbf{X}^\top \tilde{\mathbf{y}} \\ &= \mathbf{X}(\mathbf{G}\mathbf{X}^\top \tilde{\mathbf{y}}) \end{aligned}$$

(Theorem 5 in Matrices), which is in $\mathcal{C}(\mathbf{X})$ (Definition 9 in Subspaces and Rank). For $\tilde{\mathbf{y}} - \tilde{\mathbf{u}}$,

$$\begin{aligned} \mathbf{X}^\top (\tilde{\mathbf{y}} - \tilde{\mathbf{u}}) &= \mathbf{X}^\top \tilde{\mathbf{y}} - \mathbf{X}^\top \tilde{\mathbf{u}} && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1\text{)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - \mathbf{X}^\top (\mathbf{X}\mathbf{G}\mathbf{X}^\top \tilde{\mathbf{y}}) && \text{(substitute } \tilde{\mathbf{u}}\text{)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \tilde{\mathbf{y}}) && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) \tilde{\mathbf{z}} && \text{(first step)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - ((\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X})) \tilde{\mathbf{z}} && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - (\mathbf{X}^\top \mathbf{X}) \tilde{\mathbf{z}} && \text{(Definition 0.4)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - \mathbf{X}^\top \tilde{\mathbf{y}} && \text{(first step)} \\ &= \tilde{\mathbf{0}}_p, && \text{(arithmetic)} \end{aligned}$$

so $\tilde{\mathbf{y}} - \tilde{\mathbf{u}} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ (Theorem 30 in Subspaces and Rank). By the uniqueness in Theorem 31 in Subspaces and Rank, $\tilde{\mathbf{u}}$ is the orthogonal projection of $\tilde{\mathbf{y}}$ onto $\mathcal{C}(\mathbf{X})$.

Same matrix. Every choice of $\mathbf{G}$ gives a matrix that projects every $\tilde{\mathbf{y}}$ onto $\mathcal{C}(\mathbf{X})$, so all choices give the same matrix (Theorem 0.5, part 2). When $\text{rank}(\mathbf{X}) = p$, $(\mathbf{X}^\top \mathbf{X})^{-1}$ is one choice (Theorem 0.1, Theorem 0.8), and it gives $\mathbf{H}$. $\square$

Example 0.22 (A design matrix with a repeated column). Let $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, an intercept column entered twice, so $\text{rank}(\mathbf{X}) = 1 < 2$ and $\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ is not invertible: its rank

is $\text{rank}(\mathbf{X}) = 1 < 2$ ([Theorem 19 in Subspaces and Rank](#)), while an invertible 2×2 matrix has rank 2 ([Example 0.21](#)). The matrix $\mathbf{G} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{bmatrix}$ is a generalized inverse of it:

$$\begin{aligned} (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) &= \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \quad (\text{Definition 7 in Matrices, for the first product}) \\ &= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}. \quad (\text{Definition 7 in Matrices}) \end{aligned}$$

Then

$$\begin{aligned} \mathbf{X} \mathbf{G} \mathbf{X}^\top &= \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad (\text{Definition 7 in Matrices, for } \mathbf{X} \mathbf{G}) \\ &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad (\text{Definition 7 in Matrices}) \end{aligned}$$

the hat matrix of the intercept-only model ([Example 0.2](#)). Another generalized inverse, $\mathbf{G}' = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$, which is one because $(\mathbf{X}^\top \mathbf{X}) \mathbf{G}' = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ and that times $\mathbf{X}^\top \mathbf{X}$ is again $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, gives $\mathbf{X} \mathbf{G}' = \begin{bmatrix} 0 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$ and the same $\mathbf{X} \mathbf{G}' \mathbf{X}^\top = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$, as [Theorem 0.15](#) says it must. Both project onto $\mathcal{C}(\mathbf{X}) = \text{span}\{(1, 1)\}$, the column space of the intercept-only design.

Solving linear systems and least squares

Source

This section is adapted from [Zhou \(2024b\)](#) and [Zhou \(2024a\)](#), used under the MIT License (see the license text in [Section 2 in Subspaces and Rank](#)). The first source states that every invertible matrix has an LU factorization; that is false without reordering rows (the 2×2 row swap $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has none), so this section states only how an LU factorization, when one exists, solves a system. The least squares results are proved here from the closest-point theorem ([Theorem 0.3](#)) rather than by differentiating, and the QR factorization is derived from Gram-Schmidt ([Theorem 9 in Inner Products and Orthogonality](#)).

Definition 0.7 (Triangular matrix). A square matrix $\mathbf{U}$ is **upper triangular** if every entry below the diagonal is 0: $u_{ij} = 0$ whenever $i > j$. A square matrix $\mathbf{L}$ is **lower triangular** if every entry above the diagonal is 0: $\ell_{ij} = 0$ whenever $i < j$. A lower

triangular matrix whose diagonal entries are all 1 is **unit lower triangular**.

Example 0.23 (Triangular and not).

- $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$ is upper triangular: its entries below the diagonal, in positions $(2, 1)$, $(3, 1)$ and $(3, 2)$, are 0.
- $\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$ is unit lower triangular.
- A diagonal matrix ([Definition 4 in Special Matrices and Decompositions](#)) is both upper and lower triangular.
- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is neither: its $(2, 1)$ entry is not 0, and neither is its $(1, 2)$ entry.

Theorem 0.16 (Solving a triangular system by substitution). *Let $\mathbf{U}$ be an $n \times n$ upper triangular matrix ([Definition 0.7](#)) whose diagonal entries $u_{11}, \dots, u_{nn}$ are all nonzero. For every $\tilde{\mathbf{b}} \in \mathbb{R}^n$, the system $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ has exactly one solution, given for $i = n, n-1, \dots, 1$ in turn by*

$$x_i = \frac{1}{u_{ii}} \left(b_i - \sum_{j=i+1}^n u_{ij} x_j \right)$$

*(back substitution). Likewise, if $\mathbf{L}$ is lower triangular with nonzero diagonal, $\mathbf{L}\tilde{\mathbf{y}} = \tilde{\mathbf{b}}$ has exactly one solution, given for $i = 1, \dots, n$ by $y_i = \frac{1}{\ell_{ii}} \left(b_i - \sum_{j=1}^{i-1} \ell_{ij} y_j \right)$ (**forward substitution**).*

Proof

Proof. Row i of $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ reads

$$\begin{aligned} b_i &= \sum_{j=1}^n u_{ij} x_j && \text{(Definition 11 in Matrices)} \\ &= \sum_{j=i}^n u_{ij} x_j && (u_{ij} = 0 \text{ for } j < i) \\ &= u_{ii} x_i + \sum_{j=i+1}^n u_{ij} x_j. && \text{(split off the } j = i \text{ term)} \end{aligned}$$

This equation is equivalent to each of

$$u_{ii} x_i = b_i - \sum_{j=i+1}^n u_{ij} x_j \quad (\text{subtract the sum from both sides})$$

$$x_i = \frac{1}{u_{ii}} \left(b_i - \sum_{j=i+1}^n u_{ij} x_j \right). \quad (\text{divide by } u_{ii} \neq 0)$$

Row n involves only x_n , so it fixes x_n ; once $x_{i+1}, \dots, x_n$ are fixed, row i fixes x_i . So the rows, taken from the last to the first, hold exactly when $\tilde{x}$ is the vector the formula builds: there is one solution, and only one. For $\mathbf{L}$, row i reads $\sum_{j=1}^{i-1} \ell_{ij} y_j + \ell_{ii} y_i = b_i$, and the same argument runs from the first row to the last. $\square$

Example 0.24 (Back substitution on a 3×3 system). Solve $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix} \tilde{x} = \begin{bmatrix} 8 \\ 1 \\ 1 \end{bmatrix}$:

1.

$$\begin{aligned} x_3 &= \frac{1}{-1} \cdot 1 \\ &= -1; \end{aligned}$$

2.

$$\begin{aligned} x_2 &= \frac{1}{1/2} \left(1 - \frac{1}{2} \cdot (-1) \right) \\ &= 2 \cdot \frac{3}{2} \\ &= 3; \end{aligned}$$

3.

$$\begin{aligned} x_1 &= \frac{1}{2} (8 - 1 \cdot 3 - (-1)(-1)) \\ &= \frac{1}{2} \cdot 4 \\ &= 2. \end{aligned}$$

So $\tilde{x} = (2, 3, -1)$. Without a nonzero diagonal the method fails: $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \tilde{x} = (1, 1)$ has no solution, since its second row reads $0 = 1$.

Theorem 0.17 (QR factorization). *Let $\mathbf{A}$ be an $m \times n$ matrix with $\text{rank}(\mathbf{A}) = n$. Then*

$$\underbrace{\mathbf{A}}_{m \times n} = \underbrace{\mathbf{Q}}_{m \times n} \underbrace{\mathbf{R}}_{n \times n},$$

where the columns of $\mathbf{Q}$ are orthonormal ([Definition 17 in Vectors](#)) and $\mathbf{R}$ is upper

triangular (Definition 0.7) with positive diagonal entries.

i Proof

Proof. The n columns $\tilde{a}_1, \dots, \tilde{a}_n$ of $\mathbf{A}$ are linearly independent (Definition 3 in Subspaces and Rank), so the Gram-Schmidt process (Definition 10 in Inner Products and Orthogonality) runs all n steps and gives orthonormal $\tilde{q}_1, \dots, \tilde{q}_n$ (Theorem 9 in Inner Products and Orthogonality, parts 1 and 3); let $\mathbf{Q}$ have these columns. At step i , $\tilde{q}_i \neq \tilde{0}$, and rearranging the orthogonalize step,

$$\begin{aligned}\tilde{a}_i &= \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j && \text{(add the sum to both sides of the orthogonalize step)} \\ &= \|\tilde{q}_i\| \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j. && \text{(normalize step: } \tilde{q}_i = \|\tilde{q}_i\| \tilde{q}_i \text{)}\end{aligned}$$

Let $\mathbf{R}$ be the $n \times n$ matrix with $r_{ji} = \tilde{q}_j \cdot \tilde{a}_i$ for $j < i$, $r_{ii} = \|\tilde{q}_i\| > 0$, and $r_{ji} = 0$ for $j > i$; it is upper triangular with positive diagonal. Column i of $\mathbf{QR}$ is $\mathbf{Q}$ times column i of $\mathbf{R}$ (Definition 7 in Matrices), and

$$\begin{aligned}\mathbf{Q}(r_{1i}, \dots, r_{ni}) &= \sum_{j=1}^n r_{ji} \tilde{q}_j && \text{(Theorem 3 in Subspaces and Rank)} \\ &= \sum_{j=1}^i r_{ji} \tilde{q}_j && (r_{ji} = 0 \text{ for } j > i) \\ &= r_{ii} \tilde{q}_i + \sum_{j=1}^{i-1} r_{ji} \tilde{q}_j && \text{(split off the } j = i \text{ term)} \\ &= \|\tilde{q}_i\| \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j && \text{(the entries of } \mathbf{R} \text{)} \\ &= \tilde{a}_i. && \text{(the display)}\end{aligned}$$

So $\mathbf{QR} = \mathbf{A}$. □

Example 0.25 (A QR factorization from Gram-Schmidt). For $\mathbf{A}$ with columns $\tilde{a}_1 = (1, 1, 0)$, $\tilde{a}_2 = (1, 0, 1)$, $\tilde{a}_3 = (0, 1, 1)$, Example 21 in Inner Products and Orthogonality and Example 23 in Inner Products and Orthogonality found $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$, $\tilde{q}_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$, $\tilde{q}_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)$, with $\|\tilde{q}_1\| = \sqrt{2}$, $\|\tilde{q}_2\| = \sqrt{3/2}$, $\|\tilde{q}_3\| = 2/\sqrt{3}$,

$$\begin{aligned}\tilde{q}_1 \cdot \tilde{a}_2 &= \tilde{q}_1 \cdot \tilde{a}_3 \\ &= \frac{1}{\sqrt{2}}\end{aligned}$$

and $\tilde{q}_2 \cdot \tilde{a}_3 = \frac{1}{\sqrt{6}}$. So

$$\mathbf{R} = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}.$$

As a check on the second column,

$$\begin{aligned}\frac{1}{\sqrt{2}} \tilde{q}_1 + \sqrt{3/2} \tilde{q}_2 &= \frac{1}{2} (1, 1, 0) + \frac{1}{2} (1, -1, 2) \\ &= (1, 0, 1) \\ &= \tilde{a}_2.\end{aligned}$$

Definition 0.8 (Least squares solution). Let $\mathbf{A}$ be $m \times n$ and $\tilde{\mathbf{b}} \in \mathbb{R}^m$. A **least squares solution** of $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ is a vector $\hat{\tilde{\mathbf{x}}} \in \mathbb{R}^n$ such that

$$\|\tilde{\mathbf{b}} - \mathbf{A}\hat{\tilde{\mathbf{x}}}\| \leq \|\tilde{\mathbf{b}} - \mathbf{A}\tilde{\mathbf{x}}\| \quad \text{for every } \tilde{\mathbf{x}} \in \mathbb{R}^n.$$

The equations $\mathbf{A}^\top \mathbf{A}\tilde{\mathbf{x}} = \mathbf{A}^\top \tilde{\mathbf{b}}$ are the **normal equations**.

Example 0.26 (Fitting a constant). Let $\mathbf{A} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\tilde{\mathbf{b}} = (1, 3)$. The system $x(1, 1) = (1, 3)$ is inconsistent (Definition 0.6): its first entry needs $x = 1$ and its second needs $x = 3$. For any number x ,

$$\begin{aligned}\|\tilde{\mathbf{b}} - \mathbf{A}x\|^2 &= (1 - x)^2 + (3 - x)^2 \\ &= 2(x - 2)^2 + 2,\end{aligned}$$

which is smallest at $x = 2$, so $\hat{x} = 2$ is the least squares solution. The value $x = 1$ is not one: it gives $0 + 4 = 4 > 2$. The normal equations read $2x = 4$, and their solution is $x = 2$ too.

Theorem 0.18 (Least squares solutions solve the normal equations). Let $\mathbf{A}$ be $m \times n$ and $\tilde{\mathbf{b}} \in \mathbb{R}^m$.

1. $\hat{\tilde{\mathbf{x}}}$ is a least squares solution (Definition 0.8) exactly when it solves the normal equations $\mathbf{A}^\top \mathbf{A}\hat{\tilde{\mathbf{x}}} = \mathbf{A}^\top \tilde{\mathbf{b}}$.
2. The normal equations are consistent (Definition 0.6).

3. Every least squares solution gives the same fitted vector $\mathbf{A}\hat{\tilde{x}}$: the orthogonal projection of $\tilde{b}$ onto $\mathcal{C}(\mathbf{A})$ (Definition 0.3).
4. The least squares solution is unique exactly when $\text{rank}(\mathbf{A}) = n$, and then $\hat{\tilde{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \tilde{b}$.

i Proof

Proof. Let $\tilde{u}$ be the orthogonal projection of $\tilde{b}$ onto $\mathcal{C}(\mathbf{A})$. The vectors $\mathbf{A}\tilde{x}$, as $\tilde{x}$ ranges over $\mathbb{R}^n$, are exactly the points of $\mathcal{C}(\mathbf{A})$ (Definition 9 in Subspaces and Rank). **Least squares solutions are the solutions of $\mathbf{A}\tilde{x} = \tilde{u}$.** If $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$, then for every $\tilde{x}$, $\mathbf{A}\tilde{x} \in \mathcal{C}(\mathbf{A})$, so $\|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\| = \|\tilde{b} - \tilde{u}\| \leq \|\tilde{b} - \mathbf{A}\tilde{x}\|$ (Theorem 0.3), and $\hat{\tilde{x}}$ is a least squares solution. Conversely, let $\hat{\tilde{x}}$ be a least squares solution. $\tilde{u} \in \mathcal{C}(\mathbf{A})$, so $\tilde{u} = \mathbf{A}\tilde{x}_0$ for some $\tilde{x}_0$ (Definition 9 in Subspaces and Rank), and

$$\begin{aligned} \|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\| &\leq \|\tilde{b} - \mathbf{A}\tilde{x}_0\| \quad (\text{Definition 0.8}) \\ &= \|\tilde{b} - \tilde{u}\| \quad (\tilde{u} = \mathbf{A}\tilde{x}_0) \\ &\leq \|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\|, \quad (\text{Theorem 0.3, with } \tilde{w} = \mathbf{A}\hat{\tilde{x}} \in \mathcal{C}(\mathbf{A})) \end{aligned}$$

so the last inequality is an equality, and the equality case of Theorem 0.3 gives $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$. **Parts 1 and 3.** $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$ holds exactly when $\tilde{b} - \mathbf{A}\hat{\tilde{x}} \in \mathcal{C}(\mathbf{A})^\perp$: if $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$, then $\tilde{b} - \tilde{u} \in \mathcal{C}(\mathbf{A})^\perp$ (Definition 0.3); conversely, if $\tilde{b} - \mathbf{A}\hat{\tilde{x}} \in \mathcal{C}(\mathbf{A})^\perp$, then $\tilde{b} = \mathbf{A}\hat{\tilde{x}} + (\tilde{b} - \mathbf{A}\hat{\tilde{x}})$ is a split into $\mathcal{C}(\mathbf{A})$ and $\mathcal{C}(\mathbf{A})^\perp$, so $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$ by the uniqueness in Theorem 31 in Subspaces and Rank. And $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$ (Theorem 30 in Subspaces and Rank), where

$$\begin{aligned} \mathbf{A}^\top (\tilde{b} - \mathbf{A}\hat{\tilde{x}}) &= \mathbf{A}^\top \tilde{b} - \mathbf{A}^\top (\mathbf{A}\hat{\tilde{x}}) \quad (\text{Theorem 0.7}) \\ &= \mathbf{A}^\top \tilde{b} - \mathbf{A}^\top \mathbf{A}\hat{\tilde{x}}, \quad (\text{Theorem 5 in Matrices}) \end{aligned}$$

which is $\tilde{0}_n$ exactly when the normal equations hold.

Part 2. The vector $\tilde{x}_0$ above satisfies $\mathbf{A}\tilde{x}_0 = \tilde{u}$, so it is a least squares solution, and it solves the normal equations by part 1.

Part 4. The least squares solutions are exactly the solutions of $\mathbf{A}\tilde{x} = \tilde{u}$, a consistent system, so there is exactly one when $\text{rank}(\mathbf{A}) = n$ and more than one otherwise (Corollary 0.1). When $\text{rank}(\mathbf{A}) = n$, $\mathbf{A}^\top \mathbf{A}$ is invertible (Theorem 0.1), and with $\mathbf{M} \stackrel{\text{def}}{=} \mathbf{A}^\top \mathbf{A}$,

$$\begin{aligned} \hat{\tilde{x}} &= \mathbf{I}_n \hat{\tilde{x}} \quad (\text{Theorem 7 in Matrices}) \\ &= (\mathbf{M}^{-1} \mathbf{M}) \hat{\tilde{x}} \quad (\text{Definition 5 in Special Matrices and Decompositions}) \\ &= \mathbf{M}^{-1} (\mathbf{M}\hat{\tilde{x}}) \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{M}^{-1} \mathbf{A}^\top \tilde{b}. \quad (\text{the normal equations}) \end{aligned}$$

□

Example 0.27 (A least squares line). Fit $y = x_1 + x_2 t$ to the points $(t, y) = (1, 1), (2, 2), (3, 2)$: $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ (the $\mathbf{X}$ of Example 0.1) and $\tilde{\mathbf{b}} = (1, 2, 2)$. $\mathbf{A}$ has rank 2 (Example 2 in Subspaces and Rank), so the solution is unique (part 4). $\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$ has inverse $\frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$ (Example 0.1), and

$$\begin{aligned} \mathbf{A}^\top \tilde{\mathbf{b}} &= (1 + 2 + 2, 1 + 4 + 6) \\ &= (5, 11), \end{aligned}$$

so

$$\begin{aligned} \hat{\mathbf{x}} &= \frac{1}{6} (14 \cdot 5 - 6 \cdot 11, -6 \cdot 5 + 3 \cdot 11) \\ &= \frac{1}{6} (4, 3) \\ &= \left(\frac{2}{3}, \frac{1}{2}\right). \end{aligned}$$

The fitted values are $\mathbf{A}\hat{\mathbf{x}} = (\frac{7}{6}, \frac{5}{3}, \frac{13}{6})$, and the residuals $\tilde{\mathbf{b}} - \mathbf{A}\hat{\mathbf{x}} = (-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6})$ are orthogonal to both columns of $\mathbf{A}$: $-\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = 0$ and $-\frac{1}{6} + \frac{2}{3} - \frac{1}{2} = 0$.

Example 0.28 (Many least squares solutions, one fitted vector). With the rank-1 matrix

$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$ (Example 2 in Subspaces and Rank) and $\tilde{\mathbf{b}} = (1, 2, 2)$, $\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix}$ and

$\mathbf{A}^\top \tilde{\mathbf{b}} = (5, 10)$, so both normal equations say $3x_1 + 6x_2 = 5$, that is, $x_1 + 2x_2 = \frac{5}{3}$. There are infinitely many least squares solutions, such as $(\frac{5}{3}, 0)$ and $(0, \frac{5}{6})$, as part 4 predicts for $\text{rank}(\mathbf{A}) = 1 < 2$; but every one gives the same fitted vector

$$\begin{aligned} \mathbf{A}\hat{\mathbf{x}} &= (x_1 + 2x_2) (1, 1, 1) \\ &= \frac{5}{3} (1, 1, 1), \end{aligned}$$

as part 3 says.

Theorem 0.19 (Solving least squares by QR). *Let $\mathbf{A}$ be $m \times n$ with $\text{rank}(\mathbf{A}) = n$, with QR factorization $\mathbf{A} = \mathbf{Q}\mathbf{R}$ (Theorem 0.17), and let $\tilde{\mathbf{b}} \in \mathbb{R}^m$. The least squares solution (Definition 0.8) is the unique solution of*

$$\underset{n \times n}{\mathbf{R}} \hat{\tilde{x}} = \underset{n \times m}{\mathbf{Q}^\top} \tilde{b},$$

which back substitution computes (Theorem 0.16). When $m = n$, it is the unique solution of $\mathbf{A}\tilde{x} = \tilde{b}$.

i Proof

Proof. $\mathbf{R}$ is upper triangular with positive diagonal, so $\mathbf{R}\tilde{x} = \mathbf{Q}^\top \tilde{b}$ has exactly one solution $\hat{\tilde{x}}$ (Theorem 0.16). $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n$, as in the proof of Theorem 0.4. Then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} \hat{\tilde{x}} &= (\mathbf{QR})^\top (\mathbf{QR}) \hat{\tilde{x}} && \text{(substitute } \mathbf{A} = \mathbf{QR}) \\ &= \mathbf{R}^\top \mathbf{Q}^\top (\mathbf{QR}) \hat{\tilde{x}} && \text{(Theorem 16 in Matrices)} \\ &= \mathbf{R}^\top (\mathbf{Q}^\top \mathbf{Q}) (\mathbf{R} \hat{\tilde{x}}) && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{R}^\top \mathbf{I}_n (\mathbf{R} \hat{\tilde{x}}) && (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n) \\ &= \mathbf{R}^\top (\mathbf{R} \hat{\tilde{x}}) && \text{(Theorem 7 in Matrices)} \\ &= \mathbf{R}^\top (\mathbf{Q}^\top \tilde{b}) && (\mathbf{R} \hat{\tilde{x}} = \mathbf{Q}^\top \tilde{b}) \\ &= (\mathbf{R}^\top \mathbf{Q}^\top) \tilde{b} && \text{(Theorem 5 in Matrices)} \\ &= (\mathbf{QR})^\top \tilde{b} && \text{(Theorem 16 in Matrices)} \\ &= \mathbf{A}^\top \tilde{b}, && (\mathbf{A} = \mathbf{QR}) \end{aligned}$$

so $\hat{\tilde{x}}$ solves the normal equations and is the least squares solution (Theorem 0.18), which is unique because $\text{rank}(\mathbf{A}) = n$. When $m = n$,

$$\begin{aligned} \text{rank}(\mathbf{A}) &= n \\ &= m, \end{aligned}$$

so $\mathbf{A}\tilde{x} = \tilde{b}$ is consistent for every $\tilde{b}$ (Corollary 0.1, part 1), so its solution leaves residual $\tilde{0}$, is a least squares solution, and is therefore $\hat{\tilde{x}}$. $\square$

Example 0.29 (The least squares line by QR). For $\mathbf{A}$ and $\tilde{b}$ of Example 0.27, Gram-Schmidt gives $\tilde{q}_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$; then

$$\begin{aligned} \tilde{q}_1 \cdot (1, 2, 3) &= \frac{6}{\sqrt{3}} \\ &= 2\sqrt{3}, \\ \tilde{q}_2 &= (1, 2, 3) - 2(1, 1, 1) \\ &= (-1, 0, 1) \end{aligned}$$

with norm $\sqrt{2}$, and $\tilde{q}_2 = \frac{1}{\sqrt{2}}(-1, 0, 1)$. So

$$\begin{aligned}\mathbf{R} &= \begin{bmatrix} \sqrt{3} & 2\sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}, \\ \mathbf{Q}^\top \tilde{\mathbf{b}} &= \left(\frac{1+2+2}{\sqrt{3}}, \frac{-1+0+2}{\sqrt{2}} \right) \\ &= \left(\frac{5}{\sqrt{3}}, \frac{1}{\sqrt{2}} \right).\end{aligned}$$

Back substitution gives

$$\begin{aligned}x_2 &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \\ &= \frac{1}{2}\end{aligned}$$

and

$$\begin{aligned}x_1 &= \frac{1}{\sqrt{3}} \left(\frac{5}{\sqrt{3}} - 2\sqrt{3} \cdot \frac{1}{2} \right) \\ &= \frac{5}{3} - 1 \\ &= \frac{2}{3},\end{aligned}$$

the solution found in Example 0.27.

Definition 0.9 (LU factorization). An **LU factorization** of an $n \times n$ matrix $\mathbf{A}$ is a product $\mathbf{A} = \mathbf{L}\mathbf{U}$ with $\mathbf{L}$ unit lower triangular and $\mathbf{U}$ upper triangular (Definition 0.7).

Example 0.30 (An LU factorization of a 3×3 matrix). Let $\mathbf{A} = \begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$,

$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ -1 & 4 & 1 \end{bmatrix}$ (unit lower triangular) and $\mathbf{U} = \begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$ (upper triangular). By

Definition 7 in Matrices, entry (i, j) of $\mathbf{L}\mathbf{U}$ is $\sum_k \ell_{ik} u_{kj}$, so row i of $\mathbf{L}\mathbf{U}$ is $\sum_k \ell_{ik}$ times row k of $\mathbf{U}$. Row 1 is $1 \cdot (2, 1, -1) = (2, 1, -1)$; row 2 is $-\frac{3}{2}(2, 1, -1) + (0, \frac{1}{2}, \frac{1}{2}) = (-3, -1, 2)$, and row 3 is $-(2, 1, -1) + 4(0, \frac{1}{2}, \frac{1}{2}) + (0, 0, -1) = (-2, 1, 2)$. These are the rows of $\mathbf{A}$, so $\mathbf{L}\mathbf{U} = \mathbf{A}$ is an LU factorization. The below-diagonal entries of $\mathbf{L}$ are the negatives of the multiples of earlier rows that Gaussian elimination adds to reduce $\mathbf{A}$ to $\mathbf{U}$.

Example 0.31 (An invertible matrix with no LU factorization). $\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is invertible: $\mathbf{P}^2 = \mathbf{I}_2$, so $\mathbf{P}^{-1} = \mathbf{P}$ (Definition 5 in Special Matrices and Decompositions); but it has no LU factorization. If $\mathbf{P} = \mathbf{L}\mathbf{U}$, the $(1, 1)$ entry gives $0 = 1 \cdot u_{11}$, so $u_{11} = 0$, and then

the $(2, 1)$ entry gives

$$\begin{aligned} 1 &= \ell_{21} u_{11} \\ &= 0, \end{aligned}$$

which is impossible. Swapping the two rows first removes the obstacle: the swapped matrix is $\mathbf{I}_2 = \mathbf{I}_2 \mathbf{I}_2$.

Theorem 0.20 (Solving a system with an LU factorization). *If $\mathbf{A} = \mathbf{LU}$ is an LU factorization (Definition 0.9) and the diagonal entries of $\mathbf{U}$ are all nonzero, then for every $\tilde{\mathbf{b}} \in \mathbb{R}^n$ the system $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ has exactly one solution: solve $\mathbf{L}\tilde{\mathbf{y}} = \tilde{\mathbf{b}}$ by forward substitution, then $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{y}}$ by back substitution (Theorem 0.16).*

i Proof

Proof. $\mathbf{L}$ has diagonal entries 1 and $\mathbf{U}$ has nonzero ones, so each of the two triangular systems has exactly one solution (Theorem 0.16). If $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{x}}$ are those solutions, then

$$\begin{aligned} \mathbf{A}\tilde{\mathbf{x}} &= \mathbf{L}(\mathbf{U}\tilde{\mathbf{x}}) \\ &= \mathbf{L}\tilde{\mathbf{y}} \\ &= \tilde{\mathbf{b}} \end{aligned}$$

(Theorem 5 in Matrices), so $\tilde{\mathbf{x}}$ is a solution. Conversely, if $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$, then $\tilde{\mathbf{y}} \stackrel{\text{def}}{=} \mathbf{U}\tilde{\mathbf{x}}$ satisfies

$$\begin{aligned} \mathbf{L}\tilde{\mathbf{y}} &= \mathbf{L}(\mathbf{U}\tilde{\mathbf{x}}) \\ &= (\mathbf{LU})\tilde{\mathbf{x}} \\ &= \mathbf{A}\tilde{\mathbf{x}} \\ &= \tilde{\mathbf{b}} \end{aligned}$$

(Theorem 5 in Matrices), so $\tilde{\mathbf{y}}$ is the unique solution of that system, and $\tilde{\mathbf{x}}$ is then the unique solution of $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{y}}$. $\square$

Example 0.32 (Two triangular solves). With $\mathbf{L}$ and $\mathbf{U}$ of Example 0.30, solve $\mathbf{A}\tilde{\mathbf{x}} = (8, -11, -3)$. Forward substitution on $\mathbf{L}\tilde{\mathbf{y}} = (8, -11, -3)$ gives $y_1 = 8$,

$$\begin{aligned} y_2 &= -11 + \frac{3}{2} \cdot 8 \\ &= 1, \\ y_3 &= -3 + 8 - 4 \cdot 1 \\ &= 1. \end{aligned}$$

Back substitution on $\mathbf{U}\tilde{x} = (8, 1, 1)$ is Example 0.24, which gives $\tilde{x} = (2, 3, -1)$. As a check,

$$\begin{aligned}\mathbf{A}(2, 3, -1) &= (4 + 3 + 1, -6 - 3 - 2, -4 + 3 - 2) \\ &= (8, -11, -3).\end{aligned}$$

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